

Desmos Mastery - Day 8

7 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Desmos Theory | 7 questions

[Lesson video: Desmos Theory](#)

1. For the function below, what is $f(17) - f(-5)$?

$$f(x) = a|x - 6|^2 + b|x - 6| + 9$$

Answer: _____

2. For the function below, what is $h(-1) + h(9)$?

$$h(x) = p(x - 4)^3 + q(x - 4) + 7$$

Answer: _____

3. Constants a and b satisfy $a > b > 5$. Which pair is permitted for testing a conjecture?

- A) $a = 5, b = 4$
- B) $a = 7, b = 7$
- C) $a = 6.5, b = 6$
- D) $a = 0, b = 0$

4. Which statement must be true for all positive x and y satisfying $xy = 12$?

- A) $x + y = 7$
- B) $x^2y^2 = 144$
- C) $x = 3$
- D) $x/y = 3/4$

5. If $f(x) = a(x - 2)^2 + c$ and $f(7) = 31$, what is $f(-3)$?

Answer: _____

6. A student tests $a = 1$ and finds a proposed identity holds. Which conclusion is justified?

- A) The identity is proved.
- B) The identity holds for this tested value; more reasoning is needed for all allowed values.
- C) The identity cannot fail for any integer.
- D) The identity holds only at $a = 1$.

7. Positive x and y satisfy $2x + y = 11$. What is $4^x \cdot 2^y$?

Answer: _____

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: 0 2: 14 3: C 4: B 5: 31 6: B 7: 2048

Why each answer works

1. Answer: 0. Both inputs are 11 units from 6, so $|17 - 6| = |-5 - 6|$. Every corresponding term matches; the difference is zero for all a and b .

2. Answer: 14. The shifted inputs are -5 and 5 . Their cubes and first powers cancel in the sum, leaving $7 + 7 = 14$.

3. Answer: C. $6.5 > 6 > 5$ satisfies every strict inequality. Constants need not be integers unless the problem requires integers.

4. Answer: B. Squaring $xy = 12$ gives $x^2y^2 = 144$. A particular permitted pair, such as $(3, 4)$, cannot justify the other statements for every pair.

5. Answer: 31. The inputs 7 and -3 are each 5 units from the axis $x = 2$. The squared terms are equal, so the function values are equal.

6. Answer: B. A successful test is evidence about the chosen input, not a proof for all values. Algebraic reasoning can establish an invariant; a valid counterexample can disprove one.

7. Answer: 2048. Rewrite 4^x as 2^{2x} . The product is $2^{2x+y} = 2^{11} = 2,048$. This uses the entire given constraint instead of guessing individual values.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.